COMP4003: Theory of Computation



Lecture 1: Introduction

Dr. Zhiyuan Li

Computer Science and Technology
United International College

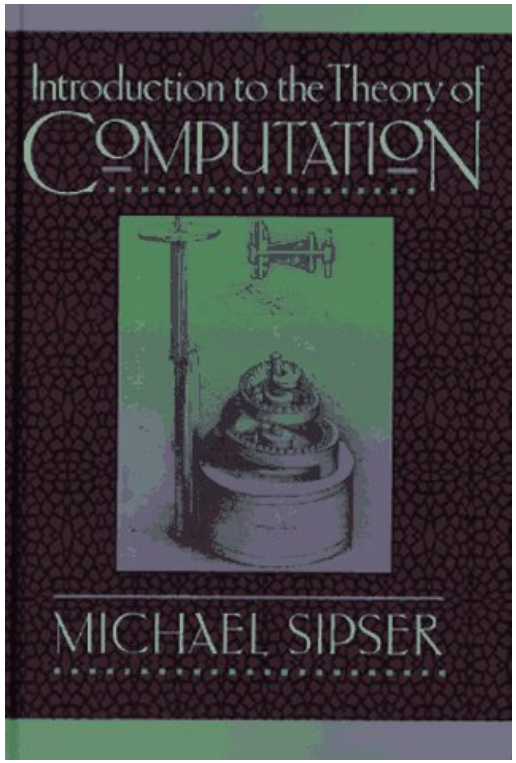
Outline

1. Course Information
   - Contact
   - Textbook
   - Reference
   - Syllabus
   - Assessment
2. Introduction to the Theory of Computation
   - Motivation
   - History

- **Instructor:** Dr. Zhiyuan Li
  - Offie: T3-502-R6
  - Email: goliathli@bnbu.edu.cn
- **TA:** Ms. Ruby Zhou
  - Office: T3-502-R26-H5
  - Email: rubinzhou@bnbu.edu.cn
- **Schedule:**
  - Lecture - Tue. 14:00-14:50 at T29-405
    Lecture - Fri. 14:00-15:50 at T29-405
  - Tutorial - TBD

Introduction to the Theory of Computation

By Michael Sipser



Reference

- On computable numbers, with an application to the Entscheidungsproblem, Alan Turing, Proceedings of the London Mathematical Society, 2(42), pp. 230-265, 1937.
- Introduction to Metamathematics, Stephen Cole Kleene, Ishi Press, 1950.
- On certain formal properties of grammars, Noam Chomsky, Information and Control, 2(2), pp. 137-167, 1959.
- Computers and Intractability: A Guide to the Theory of NP-Completemess, Michael R. Garey and David S. Johnson, New York: Freeman, 1979.
- Handbook of Theoretical Computer Science, J. van Leeuwen, The MIT Press, 1990.

Syllabus

- Introduction
- Mathematics Preliminaries
- Regular Languages
- Context-free Languages
- Recursive and Recursively Enumerable
- Computational Complexity

Assessment

Grading:

- Attendance (10%) (Goliath is trying to pass everyone.)
- Written Assignments (20%)
- Midterm Quiz (30%)
- Final Exam (40%)

Regulations:

- Late submissions are **NOT** accepted.
- Cheating will receive **ZERO** for **BOTH**.

LLM

LLMs are not avoidable in teaching and learning. Thus, students should follow these.

- The principle is "**LLM is a 24-hour teacher**".
- Teachers can answer your questions.
- Teachers cannot do assignments for you.
- Students cannot use LLMs in exams.

For example, if you want to know how to do 17×23

- you can ask your teacher "how to calculate multiplication?"
- you cannot ask your teacher "what is 17×23?"

Motivation

- What is a computation?
  - $1 + 1 =$?
  - Is 341 a prime number?
  - What is the minimum spanning tree of a given graph?
- A computation is a **procedure** to calculate the **desired output** of a **computational problem**.
- If we only solve **decision problems**,

Definition

A computation is a procedure to decide whether an element is in a set or not.

Motivation

- Are computers ultimate?
- Are all problems solveable?
- Can we write a program to test whether a program has dead loops?
- Why some problems are difficult while others are easy?
- In which case, computers are intractable?

Motivation

By studying this course, you

- can answer the above questions;
- will know the abstract models of computations;
- will know the limit of computers;
- can distinguish the difficulty of problems;
- define some new computation models (maybe); and
- (more importantly) see the ultimate truth which establishes hidden paths to connect all subjects in mathematics, computer science, logic, formal language theory, and all related majors.

Motivation

By taking this course, students can also get the following side effects.

- maximized satisfaction
- history of formalized mathematics and computer science
- write programs without using a computer
- show some tasks are not doable without implementations
- fight back when you receive contempts from students in AM, FM, STAT, etc.

Note: all of the above cannot be used in applications, only for fun.

Principle

This course is to

- study some computation models,
- compare the computational power.

When we talk about these models, they are all **ideal** with some "strange" assumptions, like

- time is infinite,
- space (memory) is infinite.

Please do not question the assumptions, just accept them.

History

Gottfried Wilhelm Leibniz (1646-1716)

- Calculus ratiocinator: a theoretical universal logical calculation framework
- Characteristica universalis: a universal conceptual language

Gottlob Frege (1848-1925)

- Begriffsschrift (Concept-script)
  - First book on symolic logic
  - To implement calculus ratiocinator

History

Georg Cantor (1845-1918)

- Set theory
- Different infinities
- Diagonal argument
- Continuum hypothesis

David Hilbert (1862-1943)

- Hilbert's program
  - All mathematics can be derived from a carefully selected finite axiom system.
  - e.g. Euclidean geometry
  - If the program had been accomplished, mathemematics would have been no more than calculus.

History

Kurt Gödel (1906-1978)

- Incompleteness theorem: if an axiom system contains natural numbers, then
  1. there is a proposition which is unprovable;
  2. and it cannot prove its consistency.
- Sabotaged Hilbert's program
- Recursion theory

Alonzo Church (1903-1995)

- Lambda calculus

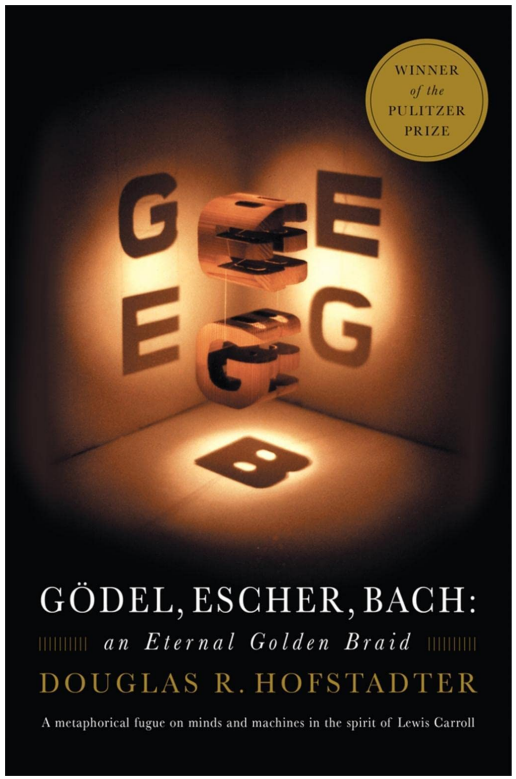
Alan Turing (1912-1954)

- Turing machine

What can you do when you can't sleep?

Gödel, Escher, Bach: an Eternal Golden Braid

By Douglas R. Hofstadter



End of Lecture 1